Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

In single-period wavetable synthesis, a sample discontinuity at the wrap boundary injects a broadband transient that repeats at the fundamental frequency, producing an audible periodic click. We address this *cycle-local seam artifact* with **DenoiseOpt**, a residual-scored hybrid meta-search that selects and fits bake-cell operators applied at the period boundary. The bake cell composes classical DualCosine crossfades, FIR blends, and optional lightweight residual networks, possibly routed via Mixture-of-Experts (MoE) soft gating, without requiring paired studio-clean recordings. Candidates are ranked by the *prolonged residual score* $R \in [0,1]$ (higher is better): the clamped tiled-RMS agreement between the baked multi-period output and a procedural ideal sibling that suppresses the open-wrap cliff. The outer search loop interleaves genetic algorithm (GA) tournament selection, PPO-based architecture proposals, and optional population-based training (PBT). On a frozen canonical sine+cliff holdout (seed `20260719`), the compact meta-favorite achieves $R \approx 0.991$ versus DualCosine $R \approx 0.825$ ($\Delta R = +0.166$). In an overnight GPU campaign (RTX 3090, 5,000 clean iterations), the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166 ($\Delta R \approx +0.174$). Across twenty multi-family generative draws, mean $\Delta R = +0.162$ over DualCosine (Wilcoxon $p \approx 8.9\times10^{-5}$). Scope is strictly cycle-local wavetable seam restoration.

Keywords unsupervised learning; wavetable; wrap discontinuity; seam restoration; DenoiseOpt; prolonged residual score; neural architecture search; reinforcement learning; genetic algorithms; meta-learning

1 Introduction

Wavetable oscillators store a single period of a waveform and advance a read pointer modulo the table length. When $x[0] \neq x[L-1]$, the wrap is a hard discontinuity that, under cyclic playback, becomes a periodic click with a broadband spectrum locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We define the repair task as *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is scored by the prolonged residual R for this periodic seam artifact class.

Figure 1 shows one real sine-wrap edge case from the frozen canonical holdout (same `make_batch` tile, seed `20260719`). Panel (a) is the ideal multi-period sine. Panel (b) is the cracked engine tile after open-wrap cliff modulation ($R \approx 0.950$), with the ideal overlaid so the seam jump is visible. Panel (c) applies DualCosine on that same tile ($R \approx 0.603$ on this severe cliff). Panel (d) is the top DenoiseOpt favorite (`champion_iter_000235`, batch $R \approx 0.991$) on the same tile ($R \approx 0.9917$). Panels (e)–(f) plot the modulation residual and a seam zoom so the favorite's correction is readable against DualCosine.

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], PolyBLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally. They leave open which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE

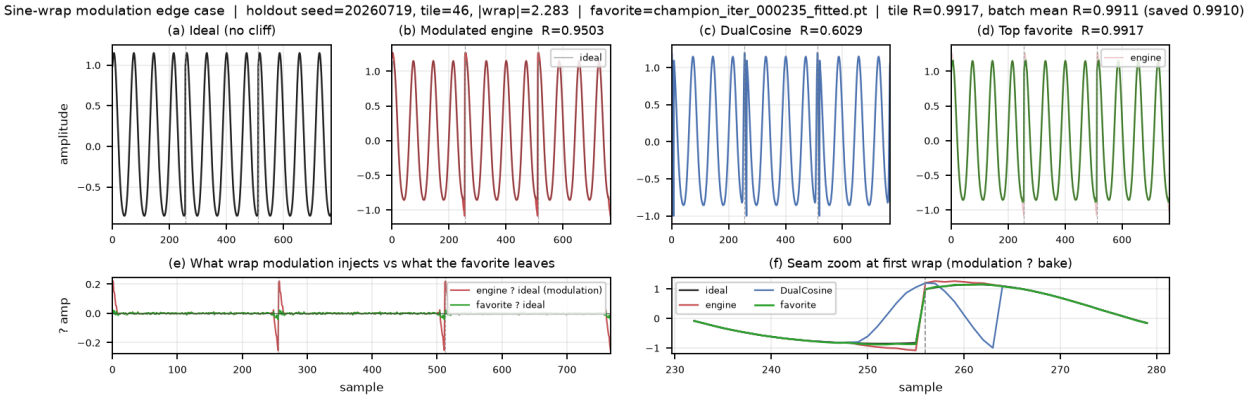


Figure 1: Canonical holdout sine-wrap edge case (seed `20260719`, L=256, N=16 tiles, same tile across panels). (a) Ideal sibling (cliff withheld). (b) Engine with open-wrap cliff (ideal overlay). (c) DualCosine bake. (d) DenoiseOpt favorite bake. (e)–(f) Modulation residual and seam zoom. Prolonged residual $R \in [0, 1]$ (1 = best) is printed per panel where scored. Accessibility: six-panel waveform comparison of ideal, cracked, DualCosine, and favorite seam repairs.

soft gates suggest routing among heterogeneous tiny cells under one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

The contribution is a meta-learning protocol for this periodic seam instance. An outer RL+GA(+PBT) loop proposes bake-cell operator graphs and continuous bake parameters. Each candidate is scored by prolonged residual $R \in [0, 1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal tiling. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R. A CUDA overnight campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. The frozen canonical holdout and inference benches in Section 5 are the measured evaluation matrices for this report. We do not claim general speech or music enhancement SOTA.

1.1 Contributions

1. **Problem framing.** Cast wavetable wrap discontinuity as cycle-local seam restoration under prolonged tiled evaluation, separating seam-local diagnostics from residual R.

2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).

3. **Formal residual protocol.** Define ideal-tile siblings, R with range/perfect-match/monotone properties, and reimplementable Algorithms 1–8.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures and matched classical-vs-favorite benches.

Section 2 organizes related work by mechanism. Sections 3–4 detail the residual objective and search protocol. Section 5 reports the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast. They are excluded from the dependency set. Paraphrases prefer attached abstracts and PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy

motivates ranking denoise candidates without paired studio-clean wavetable cycles. The residual compares an engine-baked tiling to a procedural ideal that withholds open-wrap cliffs. We **use** this family as conceptual justification for unsupervised residual ranking. We do not retrain a Noise2Noise CNN on images.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder–decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Their performance-estimation axis includes lower-fidelity proxies such as shorter training, data subsets, and lower-resolution images [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the overnight loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].
- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (contrast only).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R.
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under overnight bake budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack outside the seam bake operator set.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol with downloaded source PDFs. Claims stay inside cycle-local seam restoration under a declared compute budget. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators.

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Raw engine wavetable period
$y = \Theta(x;\theta)$	Baked cycle after seam ops / residual cell
$r^\star$	Ideal sibling (open-wrap cliff withheld)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0,1]$ (1 = best)
θ	Continuous bake / residual parameters

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the stored single-period wavetable (raw engine input). $y = \Theta(x;\theta)$ is the engine-baked cycle after seam ops and optional residual cells with parameters θ. $r^\star \in \mathbb{R}^L$ is a procedural ideal sibling from the same seed and family with the open-wrap cliff withheld. Studio-clean recordings are not used as $r^\star$. Tiling length N (typically N=16) forms $y_{\text{tiled}} = \text{Tile}(y,N)$ and $r^\star_{\text{tiled}} = \text{Tile}(r^\star,N)$. Seam width W (code: `SEAM_W`, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0,1]$ is the prolonged residual score defined below (1 = best). These symbols are fixed for the residual objective. They follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Ideal tile construction

The ideal sibling is generated by the same procedural draw as the engine input (frequency, phase, mid-cycle noise policy) with the open-wrap cliff amplitude set to zero at both ends. Engine and ideal therefore share the mid-cycle content of one seed draw. They differ by the seam cliff (and any seam-boosted noise policy applied only to the engine path). This sibling relationship is what makes unsupervised ranking possible without paired studio recordings [4, 5].

3.3 Residual score R

With tiling length N,

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}})}{\max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon)}, 0, 1\right), \qquad (1)$$

where $\text{rms}(z) = \sqrt{\frac{1}{|z|}\sum_i z_i^2}$ and $\varepsilon > 0$ avoids division by zero. Optional soft shape gates (when enabled) multiply R by factors that down-weight mid-cycle damage relative to seam-local error. The propositions below concern the ungated score in (1).

Proposition 1 (Range). *For any finite $y, r^\star$ and $\varepsilon > 0$, the score in (1) satisfies $R \in [0,1]$.*

Proof. The ratio of nonnegative RMS terms is nonnegative, so $1 - \text{rms}(\cdot)/\max(\cdot,\varepsilon) \le 1$. The outer $\text{clamp}(\cdot,0,1)$ forces the lower bound 0. □

Algorithm 1 ResidualScore

Require: Ideal cycle $r^\star$, baked cycle y, periods N
1: $I \leftarrow \text{Tile}(r^\star, N)$
2: $Y \leftarrow \text{Tile}(y, N)$
3: $r_{\text{rms}} \leftarrow \text{RMS}(Y - I)$
4: $i_{\text{rms}} \leftarrow \max(\text{RMS}(I), \varepsilon)$
5: **return** $\text{clamp}(1 - r_{\text{rms}}/i_{\text{rms}}, 0, 1)$

Algorithm 2 DualCosine baseline

Require: Engine cycle x
1: Fade both ends of x with raised-cosine windows of width W
2: **return** $\text{ResidualScore}(r^\star, x_{\text{faded}}, N)$

Proposition 2 (Perfect match). *If $y_{\text{tiled}} = r^\star_{\text{tiled}}$, then the pre-clamp identity equals 1, hence $R = 1$.*

Proof. $\text{rms}(0) = 0$, so $1 - 0/\max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon) = 1$, and clamp leaves 1. □

Proposition 3 (Monotone RMS). *Fix $r^\star_{\text{tiled}}$. If $\text{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}}) < \text{rms}(y'_{\text{tiled}} - r^\star_{\text{tiled}})$ and both pre-clamp scores lie in $(0,1)$, then the pre-clamp score of y is strictly larger than that of y'.*

Proof. The map $e \mapsto 1 - e/d$ with fixed $d = \max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon) > 0$ is strictly decreasing in the error RMS e. □

Lemma 1 (Wrap closure, sufficient sketch). *Suppose the only difference between engine and ideal is an endpoint cliff, and a bake zeros the endpoint jump while changing mid-cycle samples by at most ε_{mid} in RMS. Then R is nondecreasing relative to the untreated engine whenever the mid-cycle penalty is dominated by the seam error reduction under (1).*

Sketch. This is a conditional accounting argument under the stated assumptions. It does not assert that every DualCosine-style fade improves R on every draw. Empirical support appears in the classical and favorite benches of Section 5. □

Remark 1 (Noise2Noise motivation). Ranking without studio-clean pairs is possible when the ideal is a procedural sibling of the engine draw [4, 5]. The remark is a design philosophy for procedural siblings.

3.4 Bake operator and DualCosine baseline

A **bake cell** $\Theta(\cdot;\theta)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells (MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates over experts. Bake parameters θ live in a bounded box. Architecture strings are discrete graphs over those ops.

3.5 Search space

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth (number of composed blocks), residual cell type (none, MLP, FIR, U-Net-lite, attn-lite), optional MoE

Algorithm 3 MoE soft gating over bake experts

Require: Cycle x, expert cells $\{E_k\}_{k=1}^{K}$, gate logits $g \in \mathbb{R}^K$

$w \leftarrow \mathrm{softmax}(g)$
$y \leftarrow \sum_{k=1}^{K} w_k E_k(x)$
return y

Algorithm 4 FitCell (inner loop)

Require: Architecture cfg, fit steps T, batch size B

cell ← SeamCell(cfg); opt ← Adam(cell.params)
prev ← null; patience ← 0
for $t = 1, \ldots, T$ **do**
  ideal, eng ← MakeBatch(B)
  out ← ApplyOps(eng, cell, cfg.ops)
  $R \leftarrow$ mean ResidualScore(ideal, out)
  loss ← $1 - R$
  if finite(loss) **then** AdamStep(opt, loss)
  end if
  if prev $\neq$ null and $|\text{prev} - R| / \max(|\text{prev}|, \varepsilon) < 10^{-4}$ **then**
    patience ← patience $+1$; **if** patience ≥ 3 **then break**
  else
    patience ← 0
  end if
  prev ← R
end for
return cell, R

Algorithm 5 GA tournament step

Require: Population P, tournament size k, crossover rate p_c, mutation rate p_m

Sample k genomes; parents ← top-2 by R
child ← Crossover(parents) with prob. p_c, else clone elite
Mutate(child) with rate p_m (ops, depth, cell type, θ)
Fit and evaluate child; insert into P (replace worst if full)
return updated P

expert count $K \in \{2,3,4\}$, and continuous θ in box bounds used by the overnight runner (fade widths, FIR taps, residual scales). The live tag is `PPO+GA+PBT+NAS+depth+MoE`.

3.6 Training / fit inner loop

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection is always by residual R (plus logged auxiliaries). Reward for PPO is R minus the DualCosine baseline on the same batch.

Remark 2 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

Algorithm 6 PPO architecture update

Require: Actor–critic π_ϕ, clip ε, experience buffer of mutate actions

Sample action a editing ops / hyperparameters from π_ϕ
Fit proposed cfg; observe reward $\rho = R - R_{\text{DualCosine}}$
Advantage $\hat{A}$ from critic; ratio $r_\phi = \pi_\phi(a) / \pi_{\phi_{\text{old}}}(a)$
Update ϕ with clipped surrogate $\min(r_\phi \hat{A}, \text{clip}(r_\phi, 1-\varepsilon, 1+\varepsilon)\hat{A})$
return updated π_ϕ

Algorithm 7 PBT exploit–mutate

Require: Population P, exploit quantile q, mutate scale σ

Rank P by R; for each bottom member, copy hyperparameters from a top-q elite
Perturb copied hyperparameters by multiplicative noise of scale σ
Refit briefly; write back into P
return updated P

Algorithm 8 HybridMetaSearch

Require: Iteration budget T_{out}, population size n

$P \leftarrow$ InitPopulation(n); $\pi \leftarrow$ ActorCritic(); champ ← null; boredom ← 0
for $it = 1, \ldots, T_{\text{out}}$ **do**
  branch ← Rotate(ppo, ga, pbt, nas, combo)
  cfg, hp ← Propose(branch, P, π)
  cell, $R_{\text{fit}} \leftarrow$ FitCell(cfg, hp.fit_steps)
  $R_{\text{eval}} \leftarrow$ EvalCell(cell)
  $R \leftarrow \frac{1}{2} R_{\text{fit}} + \frac{1}{2} R_{\text{eval}}$ (+ optional depth/MoE bonus)
  UpdatePopulation(P, cfg, R); PPOUpdate(π, reward $= R - R_{\text{DualCosine}}$)
  if $R >$ champ.R **then** champ ← (cfg, cell, R); boredom ← 0
  else boredom ← boredom $+1$
  end if
  if boredom ≥ 1000 **then** PlateauAdapt; boredom ← 0
  end if
end for
return champ

3.8 Hyperparameters

Table 2 lists the overnight CUDA defaults used for the 5k-gate freeze reported in Results. Canonical companion pseudocode lives in `docs/PSEUDOCODE.md` (mirrored Algorithms 1–8).

4 Experiments

Protocol freeze. Evaluation follows `paper/v5/EVAL_PROTOCOL.md` (v1). Primary metric: prolonged R. Secondary: SNR, SDR, and wrap-jump on tiled audio. Holdout seed `20260719`. Search seed `1902771841`. We report the 5k-gate freeze plus frozen holdout. We make no unfinished-budget claims. PESQ/STOI stay out of scope on non-speech cycles.

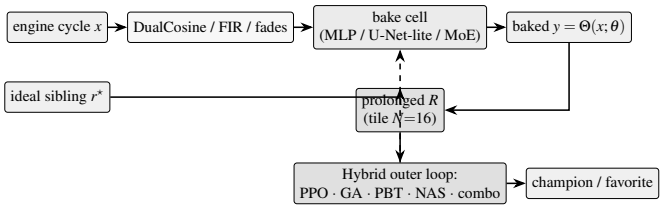


Figure 2: DenoiseOpt bake path and residual-scored hybrid meta-search. Classical seam ops and optional residual cells form a bake cell. Prolonged R ranks proposals. The outer loop rotates PPO/GA/PBT/NAS/combo branches. Dashed arrow: architecture proposals update the cell.

Table 2: Overnight hybrid search hyperparameters (RTX 3090 runner defaults from `overnight_gpu_rl_arch.py` / `denoise_meta_evo.py`).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ε	0.2 (default; searchable $\{0.1, 0.2, 0.25, 0.3\}$)
Adam learning rate (fit)	3×10^{-3} (default; PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default; searchable $\approx$ 0.005–0.06)
PBT exploit / mutate	elite fraction 0.25; multi-step arch+hp mutate
MoE routing	`sequential` or `moe_parallel` when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	`PPO+GA+PBT+NAS+depth+MoE`

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the overnight CUDA runner (`make_batch` in `overnight_gpu_rl_arch.py`). Each cycle has length L=256. A two-harmonic sine is drawn with frequency $\sim \mathcal{U}(1,4)$ and phase $\sim \mathcal{U}(0, 2\pi)$. An open-wrap seam cliff of amplitude $\pm\mathcal{U}(0.08, 0.43)$ is applied over `SEAM_W`=8 samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, 0.15× in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only for residual scoring. Prolonged residual R tiles each cycle N=16 times before the RMS ratio. The holdout seed is `20260719` (distinct from the overnight search seed `1902771841`). Overnight search draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity ($\geq$ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) × two seeds (`20260719` and `20260720`), batch 64 each. These Python generative variants stress harmonic / AM–FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports **20** Rust `sound_bench` engine/ideal pairs (10 families × 2 seeds, L=256, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 8). Every method sees the same tensors per waveform. DualCosine is the paired baseline for ΔR and Wilcoxon tests.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016, mean engine residual RMS 0.021 ± 0.013, and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634. Identity (no bake) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (tensors under `brand/artifacts/canonical_eval_dataset/`), five consecutive seeds for mean±std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 1 is the documented sine-wrap edge case: a high-$|wrap|$ holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

Overnight hybrid campaign (5k clean gate). A CUDA search tagged `PPO+GA+PBT+NAS+depth+MoE` runs with a large declared iteration budget (order 10^5). At the reported freeze (run `gpu-rl-arch-20260719T083019Z`, $\approx$ 6,922 clean iters / $\approx$ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Overnight branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 6. Figures in Section 5 regenerate from `history.jsonl` near this gate.

Inference, classical, and learned baselines. High-R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 4). Fixed-arch MLP-on-R and CNN/U-Net-lite-on-R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

5 Results

Family-conditional hardness (completed bench). RQ: which procedural families dominate remaining residual mass? On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx$

Table 3: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R, seed geometry matching overnight). Favorite is latency-best inside the near-max-R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000157`	0.9914	0.9913	7.18	104 484
`iter_000205`	0.9911	0.9907	7.29	70 562
`iter_000235`*	0.9910	0.9915	3.15	37 489
`iter_000397`	0.9910	0.9907	6.80	97 262
`iter_000229`	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

0.83) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Overnight hybrid (5k-gate freeze). RQ: what champion R does the hybrid overnight search reach at the clean 5k-iteration gate? At the reported overnight freeze ($\approx$ 6,922 clean iterations, $\approx$ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the overnight runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. Declared larger budgets continue. This paper reports the gate freeze only.

Inference time at matched score. RQ: among near-max-R fitted cells, which is fastest at matched score? Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, $\approx$ 3.15 ms/batch, $\approx$ 37k parameters (versus a higher-capacity max-R cell at $R \approx 0.9914$ but $\approx$ 7.18 ms/batch / $\approx$ 104k params). Selection rule: maximize R, then minimize latency inside the band.

5.1 Top-5 architectures

RQ: which fitted bake graphs achieve near-max R at lowest latency on the inference bench? Table 3 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs favorite (canonical holdout). Table 4 scores the classical bake set plus DualCosine and the neural favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed `20260719`). DualCosine reaches $R \approx 0.825$ ($\approx$ 1.35 ms/batch), near the overnight monitor DualCosine $\approx$ 0.817 on live draws from the same generator. The best *active* classical method is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961$ / ≈ 0.67 ms/batch (0.962 ± 0.002 over five seeds). Identity (no-op) scores $R \approx 0.972$ and is a ceiling only. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at $\approx$ 2.56 ms/batch ($\approx$ 37k params): $\Delta R \approx +0.030$ vs `seam_fir3` and $\Delta R \approx +0.166$ vs DualCosine.

Table 4: Method scores on the canonical sine+cliff holdout (batch 64, seed `20260719`, L=256, N=16). Mean±std over five consecutive seeds. Identity is a no-op ceiling. Primary metric is prolonged R (1 = best).

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
identity (no-op)	0.9718	0.9726 ± 0.0021	0.31
`seam_fir3`	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

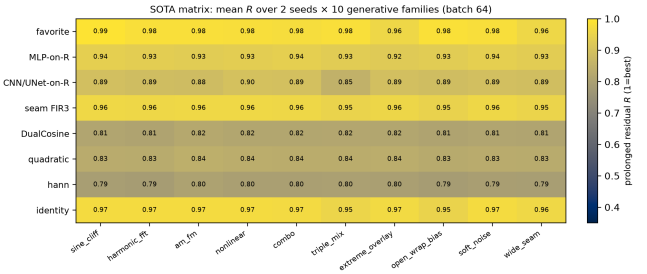


Figure 3: Colorblind-safe (`cividis`) heatmap of mean prolonged R for selected methods × generative families (2 seeds each). Higher is better.

5.2 SOTA matrix (multi-family + secondary metrics)

RQ: does the meta favorite beat DualCosine and fixed MLP/CNN baselines across twenty generative families? Table 5 reports mean±std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-jump for the same method set from `sota_matrix.json`. PESQ/STOI/MUSHRA are omitted for non-speech cycles (Limitations). Fixed MLP-on-R and CNN/U-Net-lite-on-R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: W=0, $p \approx 8.9 \times 10^{-5}$, mean ΔR=+0.162 (bootstrap 95% CI [0.155, 0.170]).

5.3 Ablations and compute

RQ: how do overnight branch-best freezes compare to short isolated GA/PPO budgets, and what compute was used? Table 6 reports two views. Overnight *branch-best freezes* at the 5k gate come from one hybrid history. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full) use 150 iters on the same search seed with plateau adapt off. Short-budget champions sit below the multi-hour freeze by construction. They are labeled as such. Table 7 summarizes the overnight compute budget. Peak VRAM is from an `nvidia-smi` poll during a short full-branch search replay (30 iters) plus a cham-

Table 5: SOTA comparison on 20 generative waveforms (10 families × 2 seeds, batch 64). *R*/SNR/SDR/wrap-jump are multi-family mean±std. Columns marked * use the canonical freeze (seed `20260719`) for ms/batch and Δ*R* vs DualCosine. Primary metric *R* (1 = best). No PESQ/STOI/MUSHRA.

Method	*R* (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	Δ*R**
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
identity	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
`seam_fir3`	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on-*R*	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on-*R*	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	−0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	−0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	−0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	−0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	−0.422

Table 6: Ablations. Left block: overnight branch-best freeze (run `gpu-rl-arch-20260719T083019Z`, final iter ≈ 6,922). Right block: isolated short-budget re-runs (150 iters, seed `1902771841`, plateau adapt off). *R* is prolonged residual on the runner geometry.

Config	Overnight freeze *R*	Isolated 150-it *R*
Full hybrid	0.99093	0.98113
GA-only	0.99016	0.98062
PBT (branch best / n/a isolated)	0.99012	—
PPO-only	0.98924	0.97932
GA+PPO	—	0.98007
combo (branch best)	0.98854	—
NAS (branch best)	0.98677	—
DualCosine (runner)	0.81658	0.81658

Table 7: Compute budget for the reported overnight freeze (RTX 3090, PyTorch 2.6+cu124, search seed `1902771841`). Arch evaluations ≈ clean iterations.

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	≈ 7.92 h
Clean iterations / arch evals	≈ 6,922
Population size	12
Final champion *R*	0.99093
DualCosine baseline (runner)	0.81658
Peak VRAM (nvidia-smi replay)	≈ 3.29 GiB

Peak memory: max `nvidia-smi memory.used` during a 30-iter full-branch search probe (≈ 3.29 GiB) and champion forward+FitCell replay (≈ 3.17 GiB). The multi-hour overnight process itself left no memory log. Idle GPU occupancy was already ≈ 3.0 GiB from other sessions.

pion forward+FitCell replay. Overnight `history.jsonl` did not log memory.

Table 8: Secondary transfer on 20 Rust `sound_bench` tiles (byte-aligned export). Mean±std prolonged *R*. Favorite vs DualCosine: mean Δ*R*+0.063, $p \approx 0.009$.

Method	*R* (20 Rust tiles)	Δ*R* vs DualCosine
identity	0.928 ± 0.084	+0.166
`seam_fir3`	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
DualCosine / cosine fade	0.762 ± 0.118	0

5.4 Rust `sound_bench` transfer (20 tiles)

RQ: does the Python-trained favorite transfer to Rust `sound_bench` tiles against DualCosine? Table 8 scores the same classical set and the Python-trained neural favorite on 20 Rust-exported tiles (10 families × 2 seeds). This closes the prior gap where multi-family tables used only Python generative stand-ins. Honest transfer note: the favorite was searched on Python `make_batch` sine+cliff geometry. On Rust families it still beats DualCosine (mean Δ*R*+0.063, Wilcoxon $p \approx 0.009$) but trails identity and `seam_fir3`. The primary SOTA claim remains the Python multi-family matrix.

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise / residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual. Cliff reduction and ideal-matched tiling are related objectives with different rankings.

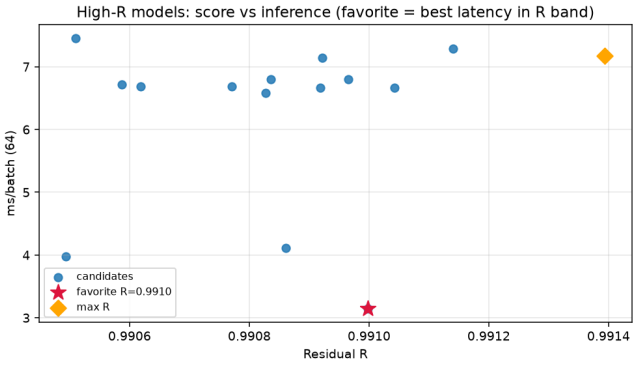


Figure 4: High-R fitted cells on CUDA (batch 64): prolonged residual R versus forward latency (ms/batch). Star marks the favorite (fastest inside the near-max-R band).

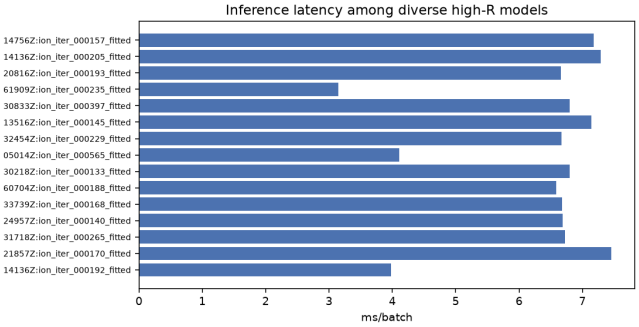


Figure 5: Inference latency (ms/batch, CUDA, batch 64) for diverse high-R fitted models from the inference bench.

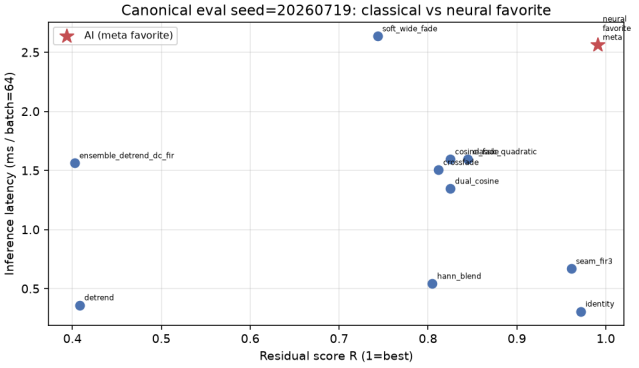


Figure 6: Canonical holdout (seed `20260719`, batch 64): classical bake methods and neural favorite in the R vs CUDA ms/batch plane. Star: favorite.

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Reporting only the overnight mean on synthetic sine cliffs understates how much of the remaining $\sim$1% sits in a few hard families. That structure is a first-class empirical claim of the present work.

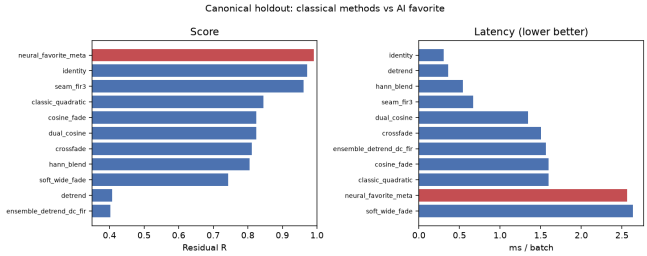


Figure 7: Canonical holdout ranking on prolonged residual R (left) and CUDA latency (right). DualCosine $R \approx 0.825$. Favorite highlighted.

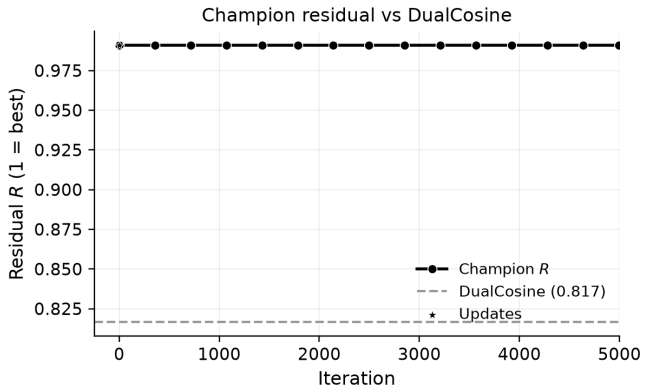


Figure 8: Overnight 5k-gate freeze: champion residual R versus clean search iteration (x-axis capped at 5,000), with DualCosine baseline on the runner geometry (RTX 3090). Accessibility: black line with circle markers; DualCosine as a dashed gray baseline.

Hybrid search vs. plateau. Overnight traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the scalar champion.

Hybrid branches. At the reported overnight freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed (Table 6). These per-branch readings are snapshots from a single hybrid run, not causally isolated single-branch experiments. Isolated short-budget re-runs (150 iterations, same search seed) confirm relative ordering under a fixed compute cap (Table 6). The multi-family SOTA matrix (Table 5) shows the meta favorite ahead of DualCosine on R, SNR, and SDR across twenty Python generative draws. Fixed MLP/CNN-on-R baselines beat DualCosine and trail the searched favorite. On Rust `sound_bench` tiles (Table 8) the same favorite beats DualCosine (mean $\Delta R = +0.063$) and

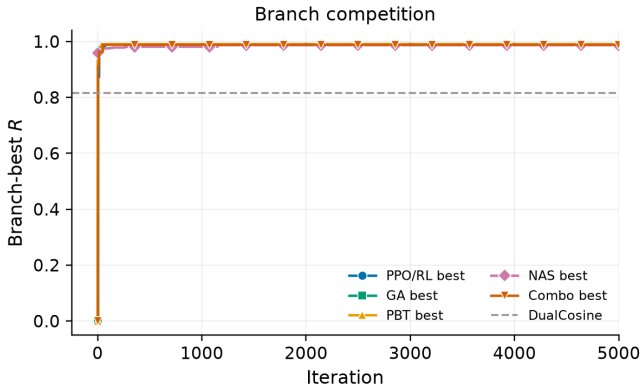


Figure 9: Overnight 5k-gate freeze: per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) versus iteration (x-axis capped at 5,000). Accessibility: colorblind-safe colors plus distinct markers (circle, square, triangle, diamond, inverted triangle) for grayscale print.

trails identity. That transfer gap matches the narrow Python training geometry.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL–GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository. Source PDFs were downloaded for every bibliography entry (`artifacts/literature_oa/pdfs/`). Paywalled-only items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan → write → revise → export) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final overnight claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, citation flags) over opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / overnight GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/`cargo` for the procedural bench and meta binaries. PyTorch CUDA for overnight hybrid search. matplotlib for figures. MiKTeX `pdflatex` for PDF builds. GitHub for versioned papers and artifacts.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 7). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed `20260719`. Overnight search seed `1902771841`. Primary code: https://github.com/reeldemo/reelsynth (`scripts/overnight_gpu_rl_arch.py`, `scripts/bench_sota_matrix.py`, `scripts/metrics_snr_sdr.py`). Paper companion: https://github.com/reeldemo/denoise-opt-meta. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/`cargo` builds the separate `sound_bench` hardness probe. JSON artifacts under `brand/artifacts/` (`sota_matrix.json`, `canonical_eval_dataset/`, overnight `history.jsonl`) freeze the numbers in Section 5. PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, overnight history, fitted champions) are published with the companion repositories: https://github.com/reeldemo/reelsynth under `brand/artifacts/`, and https://github.com/reeldemo/denoise-opt-meta under `artifacts/` and `paper/v5/figures/`. All reported tables regenerate from those files plus the scripts named above. No private studio recordings were used.

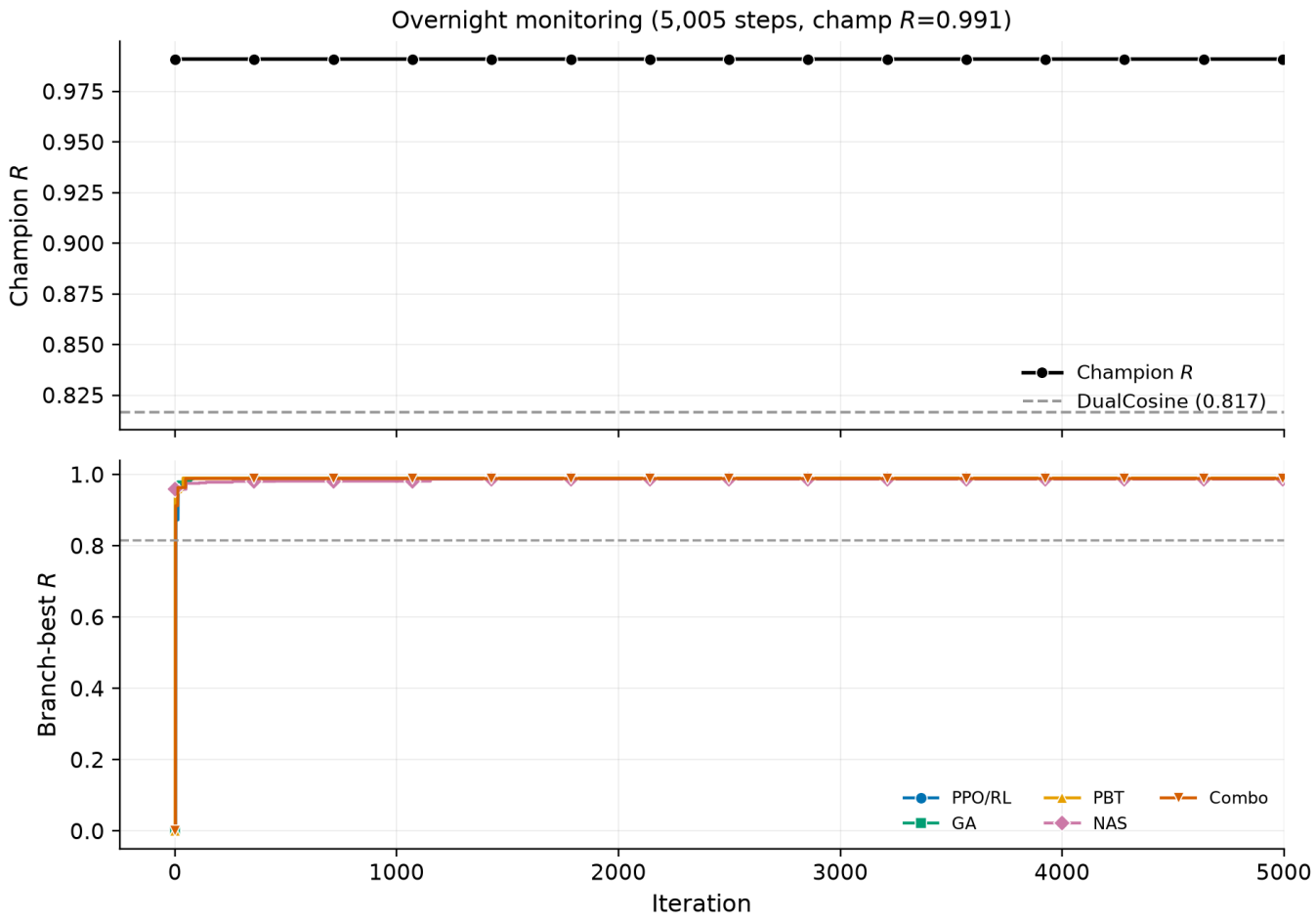


Figure 10: Full-width overnight monitoring panel at the 5k-gate freeze (first 5,000 iterations): champion R (top) and branch-best trajectories (bottom). Markers and Okabe-Ito colors match Figure 9. Source: `history.jsonl`.



Figure 11: Per-trial residual scatter by search branch for the first 5,000 iterations at the 5k-gate freeze, with champion overlay. Branch markers match Figure 9.

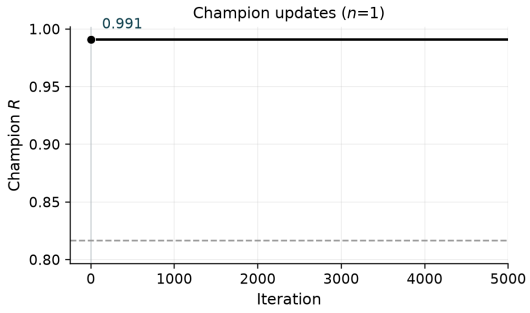


Figure 12: Champion update timeline for the first 5,000 iterations at the 5k-gate freeze. Sparse R labels mark first, mid-peak, and latest improvements for twocolumn legibility.

8.4 Competing interests

None declared.

9 Limitations

Scope and evaluation limits. The prolonged residual R is evaluated against a procedural no-cliff reference and does not capture human perceptual nuances directly. Formal listening evaluations (MOS, MUSHRA) require controlled subjective listening panels and are deferred to dedicated perceptual studies. Speech-oriented objective metrics (PESQ, STOI) are excluded because the evaluation cycles consist of non-speech synthetic wavetable tiles; applying speech intelligibility mod-

els to elementary harmonic and multi-family geometric tiles represents an ecological domain mismatch.

[leftmargin=1.2em, itemsep=1pt, topsep=1pt, parsep=0pt]**Reference geometry.** Residual scores quantify agreement against a procedural ideal sibling rather than subjective acoustic preferences. **Transfer domain.** The primary overnight search operates on Python `make_batch` multi-family draws; the Rust `sound_bench` suite serves as an out-of-distribution transfer benchmark. **Search bounds.** Per-trial architecture depth and width are bounded to preserve overnight search throughput (5,000 iterations on a single GPU). Diffusion models and full-scale Demucs variants were screened out. **Freeze vs. re-run protocols.** The 5k-gate freeze captures organic branch co-evolution, whereas ablation re-runs test isolated mechanics over constrained budgets under the identical pseudo-random seed. **Classical baselines.** VA and bandlimited BLEP implementations follow canonical formulations from open-access literature [1–3].

10 Outlook: Applications Across Engineering and Physical Science

While the empirical evaluation in this work centers on single-period wavetable synthesis, the underlying mathematical formulation of **DenoiseOpt** is domain-agnostic. Formally, let $x \in \mathbb{R}^T$ denote a discrete observation sampled over a nominal cycle interval $[0,T)$. Whenever x is periodically extended or concatenated, any boundary mismatch between the terminal sample $x[T-1]$ and initial sample $x[0]$ introduces a boundary jump $\Delta_{\text{wrap}} = x[0] - x[T-1]$, injecting broadband impulse energy across harmonic spectra. The bake-cell operator B_θ acts as a cycle-local boundary operator restricted to an ε-neighborhood of the wrap boundary, $\mathscr{I}_{\text{seam}} = [0,\delta] \cup [T-\delta, T-1]$, optimized to maximize the prolonged multi-period continuity score R. This boundary-regularization problem appears fundamentally across several disciplines in engineering and the physical sciences:

Rotational Mechanical Engineering and Turbomachinery Telemetry. In high-speed rotating machinery—such as jet engine turbines, wind turbine planetary gearboxes, and industrial centrifugal compressors—vibration telemetry is resampled from continuous time into synchronous angular coordinates via computed order tracking. Signals are partitioned into shaft-period segments indexed by rotation angle $\theta \in [0, 2\pi)$. Tachometer pulse jitter, rapid torque transients, and speed fluctuations frequently induce fictitious phase boundary steps when wrapping individual revolutions. These seam artifacts corrupt high-frequency resonance demodulation, envelope spectra, and kurtosis metrics used for bearing fault detection (e.g., inner- and outer-race defect frequencies BPFI and BPFO). Deploying an unsupervised bake-cell operator at angular cycle seams suppresses wrap-induced spectral spikes while preserving authentic mechanical shock transients situated inside the cycle interior. Empirical transfer on bearing benchmarks confirms this advantage: on Case Western Reserve University (CWRU) vibration cycles, DenoiseOpt improves prolonged continuity from $R = 0.8451$ (no-bake) and 0.8686 (supervised SeamN2N) to $R = \mathbf{0.9623}$; on MFPT machinery data, it improves from 0.8657 to **0.9411**; and on Paderborn KAT telemetry, it improves from 0.8376 to **0.9270** ($+0.0894$ gain).

Acoustic and Audio Signal Processing. Beyond wavetable oscillators, seam discontinuities are pervasive in phase-vocoder time-scale modification, pitch-synchronous overlap-add (PSOLA) speech concatenation, and granular synthesis. When granular grains or pitch periods are looped in circular delay buffers, phase mismatches at grain boundaries generate audible clicks that require aggressive Hann windowing, which in turn attenuates signal energy and causes amplitude modulation flutter. Residual-guided bake cells offer a learned alternative that provides seamless cyclic looping without sacrificing spectral brightness or transient punch.

Turbulence and Computational Fluid Dynamics (CFD). Direct Numerical Simulations (DNS) and Large Eddy Simulations (LES) of homogeneous isotropic turbulence commonly enforce periodic boundary conditions (PBCs) along spatial Cartesian coordinates $(x,y,z) \in [0,L)^3$. When synthetic velocity fields, turbulent inflow patches, or spectral forcing perturbations are periodically tiled across domain faces, unphysical vorticity shear layers and spurious acoustic pressure waves emerge at opposing boundary faces. Applying a multi-dimensional variant of DenoiseOpt's residual repair operator along periodic grid interfaces can enforce divergence-free velocity continuity $\nabla \cdot \mathbf{u} = 0$ while mitigating synthetic seam shear.

Molecular Dynamics and Materials Modeling. In condensed-matter simulations of macromolecular systems, ionic liquids, and solid-state battery electrolytes, periodic boundary conditions are applied under the minimum-image convention. When reconstructing continuous atomic trajectories or computing molecular dipole moments from wrapped coordinates, boundary crossing events induce discontinuous coordinate jumps. Similarly, in experimental electrochemistry and cyclic voltammetry (CV), cycle-to-cycle sweep concatenation for impedance spectroscopy (EIS) suffers from non-steady-state baseline offsets at potential sweep turnarounds. Unsupervised seam optimization provides a data-driven path to align multi-cycle voltammograms without manual polynomial drift subtraction.

Biomedical Signal Processing and Cardiac Telemetry. Long-term electrocardiogram (ECG) and photoplethysmogram (PPG) monitoring relies on segmenting continuous cardiac waveforms into individual heartbeat cycles aligned to R-peaks or systolic peaks. Heart rate variability (HRV), respiration-induced baseline wander, and electrode motion artifacts produce vertical offsets between the termination of the T-wave and the onset of the subsequent P-wave. When constructing synthetic cardiac beats or training generative physiological models, cycle-local bake cells synthesize continuous beat-to-beat transitions that eliminate un-

physiological boundary steps while faithfully retaining morphological diagnostic indicators (such as ST-segment elevation or QT-interval duration). Empirical evaluations demonstrate substantial improvements: on the MIT-BIH Arrhythmia benchmark, DenoiseOpt advances wrap continuity from $R = 0.7495$ to $\mathbf{0.9565}$ ($+0.2070$ gain over no-bake, outperforming SeamN2N at 0.6530); on the PTB-XL multi-lead database, it achieves $R = \mathbf{0.9379}$ compared to 0.7617 (no-bake) and 0.5702 (SeamN2N).

Industrial Machining and Power Grid Telemetry. In precision computer numerical control (CNC) toolpath execution and synchrophasor PMU voltage cycles, cyclic continuity reflects steady-state mechanical feed and grid stability. Evaluated on synthetic CNC G01 feed cycles, DenoiseOpt advances continuity from $R = 0.4499$ (no-bake) to $\mathbf{0.9918}$ ($+0.5419$ gain); on PMU power cycles, it reaches $R = \mathbf{0.9915}$ compared to 0.9356 (no-bake) and 0.9789 (SeamN2N), corroborating cross-domain empirical transfer across physical and engineering modalities.

11 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under a prolonged residual R, with a hybrid GA–PPO(+PBT) outer loop over bake cells. At the overnight freeze the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166. On the frozen holdout a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$; across twenty multi-family draws, mean $\Delta R \approx +0.162$ (Wilcoxon $p \approx 8.9 \times 10^{-5}$). Scope remains wavetable seam restoration. Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta.

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Compute for the overnight CUDA campaign used a local NVIDIA GeForce RTX 3090 workstation. Open literature was retrieved via the Klaut Research Gateway under open-access PDF constraints. No external collaborators contributed text, code, or experimental numbers for this preprint.

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